

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: _____

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

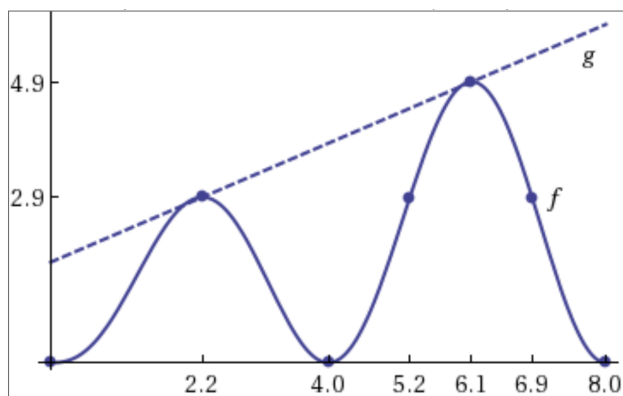
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) =$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2,) (, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x =$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x =$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

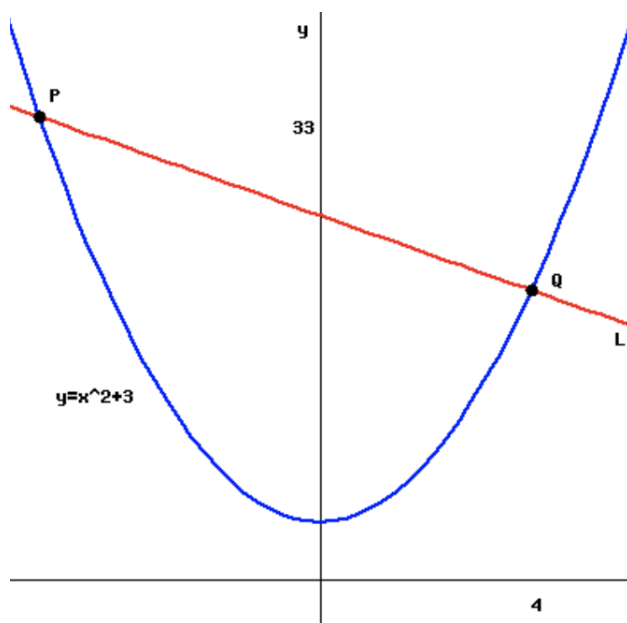
2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola (graphed in blue) has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P =$$

$$Q =$$

- (b) (4 pts) Use P and Q to determine the slope of the line L (graphed in red).

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

- (b) (2 pts) Determine the y -intercept of the parabola.

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

(b) (3 pts) $(g \circ f)(3)$

(c) (3 pts) $(f \circ f)(3)$

(d) (3 pts) $(g \circ g)(3)$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

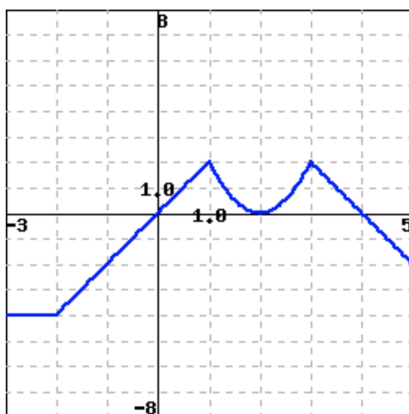
(b) (2 pts) Find $g(g^{-1}(2))$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2)$

(b) (3 pts) $w(2.5)$

(c) (3 pts) $k(1.5)$

(d) (3 pts) $n(-2.3)$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point _____

(b) (2 pts) The graph of $f(x) + 3$ must contain the point _____

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point _____

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

(b) (4 pts) $(f - g)(2)$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$